EEG Emotion Classification Report

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Analysis of 10 EEG recordings

Summary of Findings

This report contains an analysis of EEG recordings classified into 4 emotional states: Angry, Calm, Excited, Sad.

Overall classification accuracy:

- Random Forest: 0.75
- SVM: 0.50

The following pages contain detailed analysis and interpretations of the EEG data.

Data Overview

Number of recordings analyzed: 10

Distribution of emotional states:

- Angry: 2 recordings
- Calm: 3 recordings
- Excited: 3 recordings
- Sad: 2 recordings

Feature extraction method: Band powers (Delta, Theta, Alpha, Beta) and power ratios

Channels analyzed: 8 (Frontal, Central, Parietal, and Occipital regions)

Classification Results

Random Forest Classifier

True \ Predicted	Angry	Calm	Excited	Sad
Angry	1	0	0	0
Calm	0	1	0	0
Excited	1	0	0	0
Sad	0	0	0	1

SVM Classifier

True \ Predicted	Angry	Calm	Excited	Sad
Angry	1	0	0	0
Calm	0	1	0	0
Excited	1	0	0	0
Sad	0	1	0	0

Feature Importance Analysis

Random Forest Model

Feature Importance Ranking:

Feature	Importance
theta power	0.1421
beta_low power	0.1116
engagement_index	0.0981
theta_beta_ratio	0.0971
beta_mid power	0.0947
alpha power	0.0883
theta_alpha_ratio	0.0865
delta_alpha_ratio	0.0813
theta_beta_alpha_ratio	0.0757
power_ratio_index	0.0658
delta power	0.0589

Interpretation: The above features are ranked by their importance in distinguishing between different emotional states. A higher importance value indicates the feature has more influence on the classification decision.

Neurophysiological Interpretation

Interpretation of EEG patterns in emotional states:

Excited (High Arousal, High Valence)

Neurophysiological pattern: Increased Beta activity in frontal/central regions with decreased Alpha in posterior regions.

Interpretation: Beta waves (12-30 Hz) are associated with active thinking, focus, and alertness. The increased beta activity in frontal and central regions reflects the heightened engagement and alertness characteristic of excitement. Simultaneously, the suppression of alpha waves in posterior regions indicates increased sensory processing and attention to external stimuli, which is typical when someone is engaged and excited.

Angry (High Arousal, Low Valence)

Neurophysiological pattern: Increased Beta activity in right central/frontal regions.

Interpretation: The right hemisphere is more involved in processing negative emotions. Increased beta activity in the right central region (C4) approximates right frontal activity due to proximity and shared networks. This asymmetry in beta activity is consistent with the approach-withdrawal model of emotion, where negative emotions are associated with right-sided frontal activation.

Sad (Low Arousal, Low Valence)

Neurophysiological pattern: Increased Alpha activity in right posterior regions.

Interpretation: Alpha waves (8-12 Hz) are typically associated with a relaxed, disengaged state. Increased alpha in right parietal-occipital regions (PO8) during sadness suggests a withdrawal state with reduced sensory processing. This is consistent with the introspective nature of sadness, where attention is directed inward rather than to external stimuli.

Calm (Low Arousal, High Valence)

Neurophysiological pattern: Increased Alpha activity in left posterior regions.

Interpretation: The left hemisphere is more associated with positive emotions and approach behavior. Increased alpha in left parietal-occipital regions (PO7) during calmness reflects a relaxed yet positive state. This pattern supports the valence asymmetry model, with positive emotions involving relatively greater left hemisphere activation.

Conclusions and Recommendations

Key findings:

1. The classification models achieved accuracy rates of 0.75 (Random Forest) and 0.50 (SVM).

2. Distinguishing patterns were identified for each emotional state, particularly in alpha and beta frequency bands.

3. Hemispheric asymmetry plays a significant role in differentiating emotional states, supporting established theories in affective neuroscience.

Recommendations for future research:

1. Increase the sample size to improve classification reliability and allow for more complex models.

2. Incorporate additional physiological measures (e.g., ECG, GSR) for multimodal emotion classification.

3. Explore temporal dynamics of emotional states using continuous recording paradigms.

4. Validate with standardized emotion induction protocols to increase ecological validity.